TP1

1. Introduction

- **Function:** A block of code that performs a specific task.
- **Variable:** A name used to store data.
- **Method:** A function that belongs to a class.
- **Class:** A blueprint for creating objects.
- **Module:** A Python file that contains reusable code.
- **Package:** A collection of related Python modules.
- **Constant:** A value that we intend not to change.

2. Writing Code Clearly

We should write code that is easy for ourselves and other people to understand and maintain. We can do this by using meaningful names, proper formatting, and comments.
Avoid hardcoding: Instead of writing the same fixed value many times, we can store it in a variable or constant and reuse it.

Comments: Comments help explain our code. In Python, we can write comments in two common ways:

- **Single-line comment:** Uses # to explain one line or a small part of the code.
- **Multi-line comment:** We can use multiple # lines to write a longer explanation. Triple quotes (''' or """) are also used for multi-line strings and docstrings.

3. PEP 8 Rules

PEP 8 is a style guide that helps us write clean and readable Python code.

- **Indentation:** Use 4 spaces for each indentation level.
- **Line length:** Keep lines reasonably short, usually no more than 79 characters.
- **Imports:** Put import statements at the beginning of the file.
- **Naming conventions:** Use `snake_case` for variables and functions, `UPPER_CASE` for constants, and `PascalCase` for classes.

- **Whitespace:** Use spaces properly around operators and after commas to make code easier to read.

4. Debugging

Debugging means finding and fixing errors in our code. When an error occurs, we should read the error message, check the code, identify the problem, and test the code again.

5. Tips and Tricks to Help Ensure Code

- Use meaningful variable and function names.
- Avoid repeating the same code unnecessarily.
- Follow PEP 8 rules.
- Add comments when they are helpful.
- Avoid hardcoding values that may need to change.
- Test the code regularly.
- Use debugging to find and fix errors.
- Keep the code simple and easy to understand.